

Education

Master of Science in Biomedical Engineering

Arizona State University, Tempe, AZ

Bachelor of Science in Biomedical Engineering

Barrett Honors College & Arizona State University, Tempe, AZ

Professional Experience

United States Patent and Trademark Office (USPTO), VA

Jun 2025 – July 2026

Patent Examiner

- Investigate and evaluate the legal and technical merits of biomedical imaging and physiological sensing technologies, including cardiac **CT** deformation-analysis systems, depth- and point-cloud-based respiratory monitors, camera-based pulse-rate and liveness detection systems, and 4D medical image segmentation methods.
- Collaborate directly with supervisory patent examiners and prior-art search specialists to assess the novelty and non-obviousness of biomedical patent applications against domestic and international prior art.
- Conduct limitation-by-limitation claim mapping and prior-art analysis under 35 U.S.C. §§ 101, 102, 103, and 112, drafting Office Actions that determine the patentability of medical imaging and physiological sensing inventions.
- Ensure examination **quality** and consistency by **validating** evidence against primary references, identifying and correcting analytical errors before issuance, and maintaining strict compliance with USPTO documentation standards and Manual of Patent Examining Procedure guidelines.
- Analyze device **architectures**, system specifications, and signal-processing algorithms of biomedical instruments, including CT acquisition pipelines, depth-camera sensing hardware, and multi-camera optical systems, applying this technical understanding to assess novelty and enablement and support patentability determinations.

Mayo Clinic, Rochester, MN

May 2024 – May 2025

Graduate Engineer Intern

- Utilized Mimics and 3-matic Medical for segmentation of human femur CT datasets; analyzed structural integrity via stress-strain testing in ANSYS to support device-anatomy interaction modeling.
- Authored **IQ** protocol for PVD system in the Microfabrication Lab, supporting preclinical technology readiness.
- 1st Author of Feasibility study of Ti-6Al-4V additive manufacturing process characterization for point of care devices
- Drafted and implemented **SOPs** and test protocols for **software validation** (Preform) aligning with FDA and ISO guidelines. Reduced testing error by **25%**
- Led Internal Audits of Quality Management Systems, ensuring compliance with **21 CFR Part 820** and **ISO 13485**
- Collaborated cross-functionally with R&D, Quality, Regulatory, and Manufacturing Engineering teams.

Desert Platforms, Phoenix, AZ

Sep 2023 – Dec 2023

New Product Development Intern

- Designed and prototyped light-based therapeutic devices using SolidWorks and validated components through statistical analysis in **JMP**.
- Supported regulatory submissions and drafted successful proposals for **NIH Phase I SBIR** and **AFWERX** grants.

Arizona Cancer Evolution Center, Tempe, AZ

Aug 2020 – Dec 2022

Research Assistant

- Conducted statistical analysis in R to generate Phylogenetic Generalized Least Squares (PGLS) regression models.
- Received an Honorable Mention Award at the SOLUR Conference for cancer research.

Shivangi Sarkar
Cleared For Government Employment
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School of Biological and Health Systems Engineering, ASU

Jan 2022 – May 2022

Undergraduate Teaching Assistant (Jan – May 2022)

- Delivered lectures to 40+ students on transport phenomena concepts and **MATLAB** applications for BME students

Academic & Research Projects

Effects of Air Quality Index on Cognitive and Motor Scores

Jan 2025-May 2025

- Conducted statistical analysis on data from 1,397 participants to study the relationship between PM2.5 air pollution and motor performance using **R** (ggplot2, dplyr, geosphere)
- Identified a 15% improvement in motor skills among low PM2.5 exposure group; high exposure group showed no gain

Heart Disease Prediction Using Machine Learning

Jan 2024 – May 2024

- Analyzed **cardiac** datasets to identify key anatomical and physiological predictors of heart disease in **Python**.
- Applied Random Forest (92.6% accuracy) and other ML models to support virtual patient modeling.
- Conducted data segmentation and correlation analysis to simulate device-anatomy interaction.
- Strengthened understanding of **cardiac pathology** relevant to CT and fluoroscopy-based diagnosis.

Neural Signal Processing & Clustering in MATLAB

Jan 2024-May 2024

- Developed **MATLAB** scripts for segmentation and analysis of neural data to detect and classify action potentials.
- Applied PCA and k-means clustering to model neuron-level activity, simulating device-anatomy interaction and virtual signal environments.
- Visualized waveform variability using RMS, thresholding, and standard error plots to support signal integrity analysis.

Senior Capstone: Brain Cooling Non-Invasive Device

Aug 2022 – May 2023

- Designed key device components (heatsink, Peltier Cooler) using **SolidWorks**.
- Conducted heat simulations in **COMSOL** to optimize efficiency.

Technical Skills

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- Process Optimization & Validation: IQ, OQ, PQ, CAPA, GMP, FMEA, FDA (21 CFR 820), ISO 13485
 - Software & Programming: MATLAB, Python, R, SolidWorks, AutoCAD, Minitab, JMP, COMSOL, ImageJ, GraphPad, MS Suite